# PHYSICS

1) A conveyor belt is moving at a constant speed of 3 m/s. A box is gently dropped on it. The coefficient of friction between them is $\mu = 0.4$. The distance that the box will move relative to belt before coming to rest on it, is :
($g = 10\ m/s^2$)

(1) 1 m

(2) 1.125 m

(3) 2.18 m

(4) 2.42 m

2) Minimum force required on a block of weight w for motion along rough horizontal surface of coefficient of friction $\mu$ is :

(1) $\frac{\mu W}{\sqrt{1+\mu^2}}$

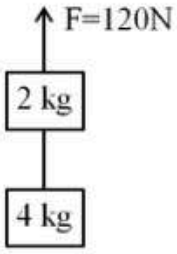
(2) $\frac{\mu W}{1+\mu}$

(3) $\frac{W}{\sqrt{1+\mu^2}}$

(4) $\sqrt{(1+\mu^2)W}$

@EXAMLIX

3) Two blocks of masses 2 kg and 4 kg are attached by an inextensible light string as shown in figure. If a force of 120 N pulls the blocks vertically upward, the tension in the string and




acceleration of the blocks is :

(1) 40 N, 10 $m/s^2$

(2) 16 N, 6 $m/s^2$

(3) 80 N, 10 $m/s^2$

(4) 20 N, 20 $m/s^2$

4) For the same weight, rolling friction is typically ..... orders of magnitude smaller than sliding friction.

(1) 1 or 2

(2) 0.5 or 1

(3) 2 or 3

(4) 3 or 4

5) The net force on a rain drop falling down with a constant speed is :

(1) Weight of the rain drop (W)

(2) Viscous drag of air (F)

(3) W + F + force of buoyancy

(4) Zero

6) Position of a body of mass 2 kg changes according to $x(t) = pt + qt^2 + rt^3$ where p = 3 m/s, q = 4 $m/s^2$ and r = 5 $m/s^3$. The force acting on the body at t = 2s is :

(1) 136 N

(2) 134 N

(3) 158 N

(4) 162 N

7) **Statement-I :** The forces of action and reaction always appear due to actual physical contact of two bodies.
**Statement-II :** A particle can move only under the action of a force.
Options :

(1) Only Statement-I is true.

(2) Only Statement-II is true.

(3) Both Statement-I and Statement-II are true.

(4) Both Statement-I and Statement-II are false.

@EXAMLIX

8) **Assertion (A) :** Newton's third law applies to all types of forces eg. gravitational force or magnetic forces etc.
**Reason (R) :** Newton's third law is applicable only when bodies are in motion.

(1) Both A and R are true and R is the correct explanation of A.

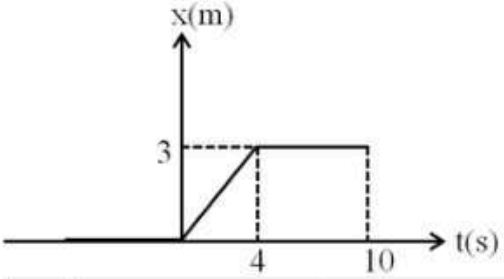
(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) Both A and R are false.

9)

Figure shows the position-time graph of a particle of mass 4 kg, then force on the particle.




| | Column-I | | Column-II |
|---|---|---|---|
| (A) | at t < 0 | (1) | –3N |
| (B) | at t > 4s | (2) | 4 N |
| (C) | 0 < t < 4s | (3) | 0 |
| (D) | from 1s to t = 5s | (4) | –0.75 N |

Options :

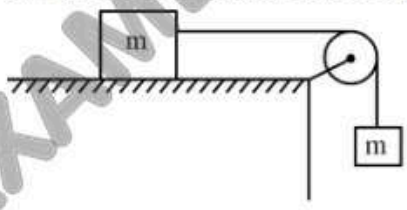
(1) A-1, B-2, C-3, D-4

(2) A-4, B-3, C-2, D-1

(3) A-3, B-3, C-3, D-4

(4) A-3, B-3, C-1, D-1

10) In the given diagram all surfaces are smooth. The additional mass required to hang so that




acceleration of the blocks becomes double.

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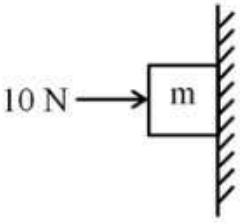
(1) m

(2) 3m

(3) 10 m

(4) infinitely large

11) A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is : ($g = 9.8\ ms^{-2}$)




(1) 0.204 kg-wt

(2) 0.24 kg-wt

(3) 2.04 kg-wt

(4) 2 kg-wt

12) **Statement-I :** If external force on a body is zero, its acceleration must be zero.

**Statement-II :** If external force on a body is zero, its velocity may be zero.

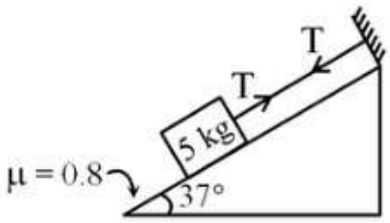
(1) Only Statement-I is true.

(2) Only Statement-II is true.

(3) Both Statement-I and Statement-II are true.

(4) Both Statement-I and Statement-II are false.

13) In the given diagram find the tension in the string :



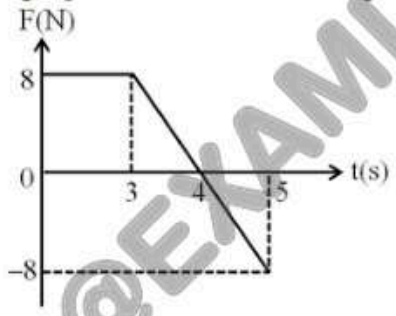

(1) 0 N

(2) 30 N

(3) 32 N

(4) 2 N

14) For the given force-time graph, calculate the change in velocity of a particle of mass 2 kg for time t = 0 to t = 4 second.




@EXAMLIX

(1) 4 m/s

(2) 14 m/s

(3) 10 m/s

(4) 28 m/s

15) A mass of 4 kg rests on a horizontal plane. The plane is gradually inclined until at an angle $\theta = 15^{\circ}$ with the horizontal, the mass just begins to slide. What is the coefficient of static friction between the block and the surface ? ($\tan 15^{\circ} = 0.27$)

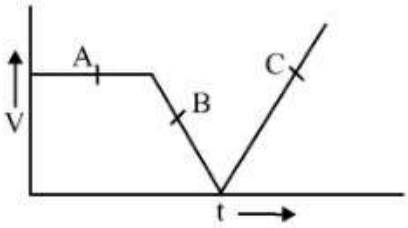
(1) 0.4

(2) 0.27

(3) 0.32

(4) 0.5

16) Velocity versus time graph of a body is shown in Fig. It explains that :-



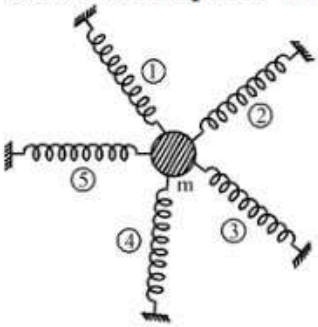

(1) At B force is zero

(2) At B there is a force but towards motion

(3) At B there is a force but opposes the motion

(4) Forces are equal at A, B and C

17) A sphere of mass 'm' is kept in equilibrium with the help of several springs as shown in the figure. Measurements show that the spring 1 applies a force $\vec{F}_1$ on the sphere. With what acceleration sphere will move immediately after this particular spring '1' is cut ?




@EXAMLIX

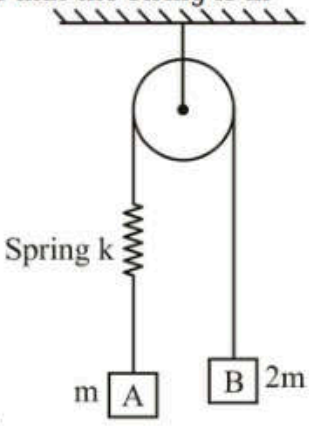
(1) zero

(2) $\frac{\vec{F}_1}{m}$

(3) $-\frac{\vec{F}_1}{m}$

(4) insufficient information

18) Two blocks A and B of masses m and 2 m respectively are held at rest such that the string is in natural length. Find out the accelerations of both the blocks just after release.



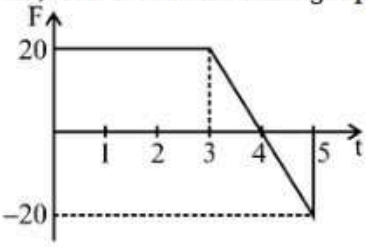

(1) $g\downarrow, g\downarrow$

(2) $\frac{g}{3}\downarrow \frac{g}{3}\uparrow$

(3) 0, 0

(4) $g\downarrow$, 0

19) For the force-time graph shown in figure the change in momentum in 5 s is :-




(1) 60 kg m/s

(2) 50 kg m/s

(3) 40 kg m/s

(4) 30 kg m/s

20) Consider the following statements and select incorrect statements :-
(A) The second law of newton is a vector law.
(B) The second law is a local law i.e, acceleration does not depend on the history of motion.
(C) The second law is applicable only for point objects.
(D) Only conservative forces are written in newton's second law.

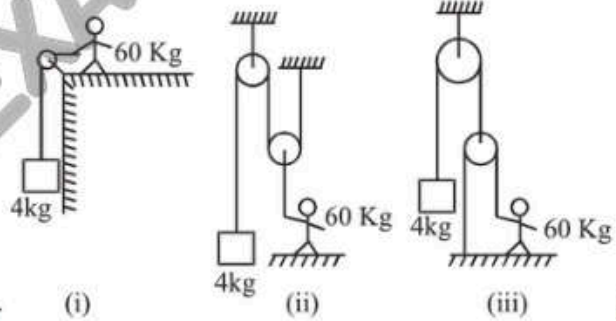
(1) Only B

(2) Only C

(3) B, D

(4) C, D

@EXAMLIX




21) Consider three cases in equilibrium :- (i) (ii) (iii) A man of mass 60 kg is holding the string connected to block of mass 4kg as shown. Then the incorrect statement is :-

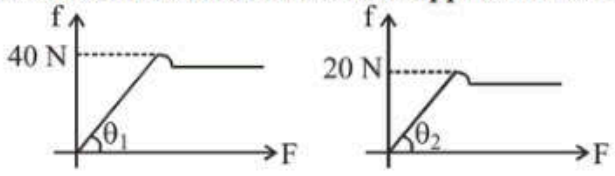
(1) Normal reaction will be maximum on man in first case

(2) Normal reaction will be least in third case.

(3) Tension in the string held by man is least in third case

(4) Tension in string held by man is maximum in second case.

22) If the two curves drawn between frictional force f and applied force F for 2 bodies on ground.




Then correct statement is :- (A) (B)

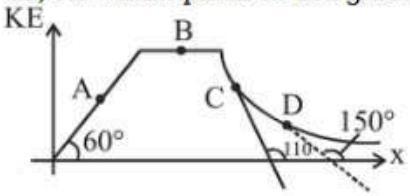
(1) $\theta_1 > \theta_2$

(2) Coefficient of static friction may be more for B than A

(3) Coefficient of kinetic friction is same as coefficient of static friction for both A & B

(4) Coefficient of static friction will be more for A than B

23) At which point in the given graph, magnitude of rate of change of momentum is maximum.




(1) A

(2) B

(3) C

(4) D

24) Which one of the following pairs of types of vectors at the same time cannot hold true for two vectors ?

(1) Collinear and concurrent

(2) Non-coplanar and concurrent

(3) Non-coplanar and collinear

(4) Concurrent and coplanar

25) Given vector $\vec{a} = 10\hat{i} - m\hat{j}$ and $|\vec{a}| = 26$, then what is the positive value of m ?

(1) 28

(2) 24

(3) 20

(4) 16

26) If the angle between two vectors $\hat{i} + \hat{k}$ and $\hat{i} - \hat{j} + \lambda\hat{k}$ is $\pi/3$, then the value of $\lambda$ is :-

(1) 0, 2

(2) 0, 4

(3) -2, 0

(4) 0, -4

27) For two non-zero vectors $\vec{A}$&$\vec{B}$, which one of the following vector operations is TRUE ?

(1) $\vec{A} \cdot \vec{B} = 1$, if $\vec{A} \perp \vec{B}$

(2) $\vec{A} \times \vec{B} = 0$, if $\vec{A} \perp \vec{B}$

(3) $\vec{A} \times \vec{B} = 0$, if $\vec{A} \| \vec{B}$

(4) $\vec{A} \cdot \vec{B} = 0$, if $\vec{A} \| \vec{B}$

@EXAMLIX

28) If a vector $\vec{A}$ making angle α, β and γ respectively with the x, y and z axes respectively then $\frac{\cos^2\alpha + \cos^2\beta + \cos^2\gamma}{\sin^2\alpha + \sin^2\beta + \sin^2\gamma} =$

(1) 0

(2) $\frac{1}{2}$

(3) $\frac{1}{3}$

(4) 1

29) If $\vec{A} = 2\hat{i} + 4\hat{j}$ and $\vec{B} = p\hat{i} + q\hat{j}$ and $\vec{A} \perp \vec{B}$, then p/q equal to :

(1) -1

(2) 1

(3) -4

(4) -2

30) Given two non zero vectors $\vec{P}$ and $\vec{Q}$ such that $|\vec{P}| > |\vec{Q}|$, then which one of the following is the possible range of the magnitude of their resultant $\vec{R}$ ?

(1) $|\vec{R}| \geq |\vec{P}| + |\vec{Q}|$

(2) $|\vec{R}| \leq |\vec{P}| - |\vec{Q}|$

(3) $|\vec{P}| - |\vec{Q}| \leq |\vec{R}| \leq |\vec{P}| + |\vec{Q}|$

(4) $|\vec{P}| - |\vec{Q}| \geq |\vec{R}| \geq |\vec{P} + \vec{Q}|$

@EXAMLIX

31) If $\vec{a}$ and $\vec{b}$ are the vectors forming consecutive sides of a regular hexagon ABCDEFA, then the vector representing the side CD is :-

(1) $\vec{a} + \vec{b}$

(2) $\vec{a} - \vec{b}$

(3) $\vec{b} - \vec{a}$

(4) $-(\vec{b} + \vec{a})$

32) If two vectors $\vec{A} = \hat{i} - \hat{k}$ and $\vec{B} = \hat{i} + \hat{j}$, then which one of the the following is false ?

(1) $\vec{A} \cdot \vec{B} = 1$

(2) $\vec{B} - \vec{A} = \hat{j} + \hat{k}$

(3) The cosine of the angle between $\vec{A}$ and $\vec{B}$ is 0.5

(4) The magnitude of $\vec{A} + \vec{B}$ is 6

33) Which of the given statement(s) is/are correct :
**Statement-I :** The magnitude of a vector is always a scalar.
**Statement-II :** Each component of a vector is always a scalar.

**Options :**

(1) Only Statement I is correct

(2) Only Statement II is correct

(3) Both Statement I and Statement II are correct

(4) Both Statement I and Statement II are incorrect

34) **Assertion :** Magnitude of resultant of two vectors may be less than the magnitude of either vector.
**Reason :** Addition and subtraction of scalars can be done only for quantities with same units.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

35) **Statement-I :** The direction of vector $\lambda\vec{A}$, $(\lambda < 0)$ same as the direction of $\vec{A}$.
**Statement-II :** If a vector is multiplied by zero, then the resultant is a scalar.

(1) Both Statement I and Statement II are false

(2) Statement I is true but Statement II is false

(3) Statement II is true but Statement I is false

(4) Both Statement I and Statement II are true

@EXAMLIX

36) If angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, then angle between $2\vec{a}$ and $-3\vec{b}$ is :-

(1) $\frac{\pi}{3}$

(2) $-\frac{\pi}{3}$

(3) $\frac{2\pi}{3}$

(4) $-\frac{2\pi}{3}$

37) Let $\vec{a}$ and $\vec{b}$ be any two vectors and $\theta$ be the angle between them, then b $\cos\theta$ is the projection of :

(1) $\vec{b}$ in the direction of $\vec{a}$

(2) $\vec{a}$ in the direction of $\vec{b}$

(3) $\vec{b}$ in the direction of x-axis

(4) $\vec{a}$ in the direction of y-axis

38) Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = |\vec{c}| = 1$, $|\vec{b}| = 4$ and $|\vec{b} \times \vec{c}| = \sqrt{15}$. If $\vec{b} - 2\vec{c} = \lambda\vec{a}$ then value of $\lambda$ is :-

(1) 1

(2) -1

(3) 2

(4) 4

39) A boat is moving in direction of vector $(-4\hat{i} + 3\hat{j})$ with a speed of 10 m/sec. Velocity vector of boat can be expressed as :-

(1) $-40\hat{i} + 30\hat{j}$

(2) $-8\hat{i} + 6\hat{j}$

(3) $-\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$

(4) $-6\hat{i} + 8\hat{j}$

40) If angle between $\hat{a}$ and $\hat{b}$ is 60° then the value of $|\hat{a} - \hat{b}|$ is :

(1) 0

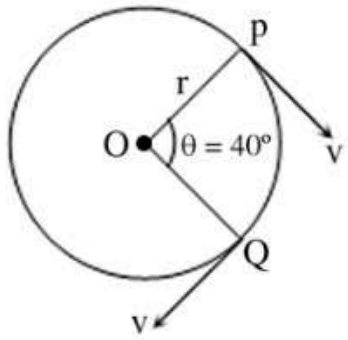
(2) 1

(3) 2

(4) 4

41)

A particle is moving on a circular path of radius r with uniform speed v. The change in velocity when the particle moves from *P* to *Q* is $(\angle POQ = 40^\circ)$




(1) $2v\cos 40^\circ$

(2) $2v\sin 40^\circ$

(3) $2v\sin 20^\circ$

(4) $2v\cos 20^\circ$

42)

If vector $\hat{i} + \hat{j} - \hat{k}$ and $2\hat{i} + 2\hat{j} + \lambda\hat{k}$ are parallel then find out the value of λ :-

(1) -1

(2) 1

@EXAMLIX

(3) -2

(4) 2

43) The component of $\vec{A} = \hat{i} + \hat{j} + 5\hat{k}$ perpendicular to $\vec{B} = 3\hat{i} + 4\hat{j}$ is

(1) $-\frac{4}{25}\hat{i} + \frac{3}{25}\hat{j} + 5\hat{k}$

(2) $-\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$

(3) $\frac{4}{25}\hat{i} - \frac{3}{25}\hat{j} + 5\hat{k}$

(4) $+\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$

44) If $\vec{A}$ & $\vec{B}$ are two vectors then which of the following are correct ?

(a) $\left(\vec{A} \times \vec{B}\right) \perp \vec{A}$

(b) $\left(\vec{A} \times \vec{B}\right) \perp \vec{B}$

(c) $\left(\vec{A} \times \vec{B}\right) \perp \left(\vec{A} + \vec{B}\right)$

(d) $\left(\vec{A} \times \vec{B}\right) \perp (\vec{A} - \vec{B})$

(e) $\left(\vec{A} \times \vec{B}\right) \perp \left(\vec{A} \cdot \vec{B}\right)$

(1) (a), (b), (c), (d), (e)

(2) (a), (b) only

(3) (a), (b), (c) only

(4) (a), (b), (c), (d) only

@EXAMLIX

45) If $\vec{P} = \vec{F}_1 + \vec{F}_2$ and $\vec{Q} = \vec{F}_1 - \vec{F}_2$. The magnitude of $F_1$ and $F_2$ are $3\sqrt{2}$N and $4\sqrt{2}$N respectively. Then the value of $\sqrt{P^2 + Q^2}$ is :

(1) Zero

(2) $6\sqrt{2}$ N

(3) 10 N

(4) $7\sqrt{2}$ N

## CHEMISTRY

1) Copper dissolved in gold is an example of :-

(1) Gas in Solid

(2) Solid in liquid

(3) Solid in Solid

(4) Liquid in Solid

2) Which of the following is/are correct for dilute solution :-
(1 → solvent, 2 → solute)
[$P_s$ = Vapour pressure of solution]

(1) $\frac{P_1^0 - P_s}{P_1^0} = \frac{w_2 M_1}{w_1 M_2}$

(2) $\frac{P_1^0 - P_s}{P_1^0} = \frac{n_1}{n_2}$

(3) $\frac{P_1^0 - P_s}{P_1^0} = \frac{n_1 - n_2}{n_1}$

(4) $\frac{P_s - P_1^o}{P_s} = \frac{n_2}{n_1}$

3) In Azeotropic mixture of nitric acid & water (% by mass) :-

(1) Nitric acid - 95%, water - 5%
(2) Nitric acid - 5%, water - 95%
(3) Nitric acid - 68%, water - 32%
(4) Nitric acid - 32%, water - 68%

4) Freezing point (F.P.) of solution containing non-volatile solute is always :-

(1) Greater than F.P. of pure solvent
(2) Less than F.P of pure solvent
(3) Equal to F.P of pure solvent
(4) May be greater or less than pure solvent

@EXAMLIX

5) In $\Delta T_b = K_b m$, $K_b$ is called :-

(1) Boiling point elevation constant
(2) Ebullioscopic constant
(3) Both (1) & (2)
(4) Cryoscopic constant

6) Unit of $K_f$ is :-

(1) $K\frac{kg}{mol}$
(2) Mole $Kg^{-1}$ $K^{-1}$
(3) Mole Kg $K^{-1}$
(4) Kg $Mole^{-1}$ $K^{-1}$

7) $K_f$ equal to :-

(1) $\frac{R\left(T_f^0\right)^2 M}{1000\Delta H_{fus}}$

(2) $\frac{R\Delta H_{fus} M}{1000\left(T_f^0\right)^2}$

(3) $\frac{\Delta H_{fus} M}{1000R\left(T_f^0\right)^2}$

(4) None of these

8) 0.6 mL of $CH_3COOH$ (density = 1.06 g/mL) is dissolved in 1 litre of water. The depression in Freezing point observed for the strength of this acid was 0.0205°C. Calculate the van't hoff factor ?

(1) 0.0197

(2) 1.041

(3) 1.2

(4) 0.0197

9) An aqueous solution of 2% non-volatile solute exerts a pressure of 1.004 bar at the normal boiling point of solvent? Calculate the molar mass of the solute ?

(1) 60.24 g $mol^{-1}$

(2) 20.24 g $mol^{-1}$

(3) 41.74 g $mol^{-1}$

(4) 30.30 g $mol^{-1}$

@EXAMLIX

10) At equilibrium, vapour phase will be rich in the component which is (take, $x_A = x_B$) :-

(1) Less volatile

(2) More volatile

(3) Whose P° is less

(4) None

11) In a solution 39 g benzene is dissolved in 46 g ethyl alcohol then the mole fraction of benzene is :

(1) 0.5

(2) 0.67

(3) 0.33

(4) 0.25

12) 2.5 g acetic acid is dissolved in 75 g benzene then molality of solution is :

(1) 0.55 m

(2) 0.43 m

(3) 0.25 m

(4) 0.4 m

13) Maximum amount of solid solute that can be dissolved in a liquid solvent does not depends on :

(1) Temperature
(2) Nature of solute
(3) Nature of solvent
(4) Pressure

14) The solubility of gas in liquid with increasing temperature :

(1) Increases
(2) Decreases
(3) Remains constant
(4) Can't predicted

15) The value of Henry's constant $K_H$ is :

(1) High for gases with higher solubility.
(2) High for gases with lower solubility.
(3) Constant for all gases.
(4) Not related to the solubility of gases.

@EXAMLIX

16) Minimum boiling azeotrope is formed by :-

(1) Phenol and Aniline
(2) Acetone and chloroform
(3) Acetone and ethanol
(4) $CHCl_3$ & $C_6H_6$

17) 18 g glucose is dissolved in 1 kg water then at what temperature this solution will be boiled? The value of $K_b$ for water is 0.52 K kg $mol^{-1}$.

(1) 373 K
(2) 373.05 K
(3) 0.052 K
(4) 373.67 K

18) Desalination of sea water is carried out by the process -

(1) Exosmosis
(2) Endosmosis
(3) Reverse osmosis
(4) Osmosis

19) Which of the following aqueous solutions will have the highest boiling point?

(1) 1 M NaOH

(2) 1 M $MgSO_4$

(3) 1 M $KNO_3$

(4) 1 M $CaCl_2$

20) From which of the following binary mixture, their components can not be seperated completely by fractional distillation.

(1) n-hexane and n-heptane

(2) $C_2H_5Br$ and $C_2H_5Cl$

(3) $C_6H_6$ and $C_6H_5$-$CH_3$

(4) $H_2O$ and $HNO_3$

21) 4 L of 0.02 M aqueous solution of NaCl is diluted with 1 L water then molarity of final solution is :

(1) 0.004 M

(2) 0.008 M

(3) 0.016 M

(4) 0.02 M

22) In comparison to a 0.01 M solution of urea, the elevation in boiling point of a 0.01 M $CaBr_2$ solution is :

(1) The same

(2) About twice

(3) About three times

(4) About six times

@EXAMLIX

23) $K_3[Fe(CN)_6]$ is 50% ionised in solution then which of the following is correct?

(1) i = 2

(2) i = 4

(3) i = 2.5

(4) i = 5.5

24) Bends occur due to toxic effect of the high concentration of -

(1) Oxygen in blood

(2) $CO_2$ in blood

(3) Nitrogen in blood

(4) He in blood

25) Which is the correct statement?

(1) Cellulose acetate is the porous membrane used for reverse osmosis

(2) Cellulose acetate is permeable to water.

(3) Cellulose acetate is impermeable to impurities and ions present in sea water.

(4) All of the above

26) Which case is valid for an ideal solution ?

(1) $\Delta_{mix}H = 0, \Delta_{mix}V = 0$

(2) $\Delta_{mix}H > 0, \Delta_{mix}V > 0$

(3) $\Delta_{mix}H < 0, \Delta_{mix}V < 0$

(4) $\Delta_{mix}H > 0, \Delta_{mix}V < 0$

27) When some CsCl was dissolved in water the freezing point depression was numerically equal to twice the molal depression constant. The relative lowering in V.P. of the solution is nearly.

(1) 0.036

(2) 0.018

(3) .0585

(4) .072

28) NaBr is 80% dissociated in aqueous solution of 0.5 m concentration. The solution freezes at ($K_f$ = 1.86 K kg $mol^{-1}$)

(1) 272 K

(2) 270 K

(3) 271.32 K

(4) 269 K

@EXAMLIX

29) Calculate the mass of $C_6H_8O_6$ to be dissolved in 75 gm of acetic acid to lower its freezing point by 1.5°C ($K_f$ = 3.9 K Kg $mol^{-1}$)

(1) 5 gm

(2) 10 gm

(3) 15 gm

(4) 2.5 gm

30) If density of sea water is 1.25 g $mL^{-1}$ and contains 92 g of $Na^+$ ions per kg of water calculate molality of $Na^+$ in sea water :-

(1) 1

(2) 2

(3) 3

(4) 4

31) $K_H$ of Methane in benzene at 298 K is $4.27 \times 10^5$ mm Hg calculate the solubility (in terms of mole fraction) of methane in benzene at 298 K and 760 mm Hg pressure :-

(1) $178 \times 10^{-5}$

(2) $250 \times 10^{-4}$

(3) $300 \times 10^{-8}$

(4) $178 \times 10^{-2}$

32) We have three solutions of KCl labelled as A, B & C with concentration 0.3 M, 0.2 M & 0.1 M respectively. The value of vant Hoff factors for three solution will be in order :-

(1) $i_A < i_B < i_C$

(2) $i_A > i_B > i_C$

(3) $i_A = i_B = i_C$

(4) $i_A < i_B > i_C$

33) One molal solution of carboxylic acid in benzene shows elevation in boiling point of 1.518 K The degree of association for dimerization of the acid in benzene is ($K_b$ for benzene is 2.53 K kg $mol^{-1}$)

(1) 75%

(2) 80%

(3) 60%

(4) 70%

34) The vant haff factor of $Hg_2Cl_2$ in its aqueous solution will be $(\alpha_{Hg_2Cl_2} = 80\%)$

(1) 1.6

(2) 2.6

(3) 3.6

(4) 4.6

@EXAMLIX

35) 18 gm glucose and 6 gm urea are dissolved in 1L aqueous solution at 27°C The osmotic pressure of the solution is :-

(1) 8.826 atm

(2) 4.926 atm

(3) 2.92 atm

(4) 4.42 atm

36) The freezing point depression of 0.001 m $K_x[Fe(CN)_6]$ is $7.1 \times 10^{-3}$ K then determine value of x ($k_f$ $H_2O$ = 1.86 k kg $mol^{-1}$)

(1) 1

(2) 2

(3) 3

(4) 4

37) At 80°C vapour pressure of pure liquid A is 520 mm Hg and that of pure liquid B is 1000 mm Hg.

If the mixture boils at 80°C and 1 atm pressure the amount of A in mixture.

(1) 50 mol %
(2) 52 mol %
(3) 34 mol %
(4) 48 mol %

38) Vapour pressure of pure liquid A & B are 450 mm & 700 mm Hg respectively at 350 K. Find out the composition of liquid mixture if total vapour pressure is 600 mm Hg ?

(1) $X_A = 0.5 ; X_B = 0.5$
(2) $X_A = 0.6 ; X_B = 0.4$
(3) $X_A = 0.4 ; X_B = 0.6$
(4) $X_A = 0.3 ; X_B = 0.7$

39) An aqueous solution of a non volatile solute boils at 100.17°C At what temperature will the solution freeze ($K_b = .512$ & $K_f = 1.86$)

(1) -0.82°C
(2) -0.62°C
(3) -0.2°C
(4) -0.4°C

@EXAMLIX

40) On mixing 10 ml $CCl_4$ with 10 ml of $C_6H_6$ the total volume of solution is :-

(1) > 20 ml
(2) < 20 ml
(3) = 20 ml
(4) = 10 ml

41) What is the molality of 10% (w/w) NaOH solution :-

(1) 5
(2) 10
(3) 2.5
(4) 2.778

42) Which has the maximum osmotic pressure at temperature T :-

(1) 100 ml of 1 M urea solution
(2) 300 ml of 1 M glucose solution
(3) Mixture of 100 ml of 1 M urea solution & 300 ml of 1 M glucose solution
(4) All are isotonic

43) Calculate the molarity of pure $CH_3OH$ having density 0.5 g/ml :-

(1) $\frac{1000}{16}$

(2) $\frac{500}{32}$

(3) $\frac{500}{16}$

(4) $\frac{1000}{64}$

44) In case of association of solute in solution, Van't Hoff factor is :

(1) 1

(2) Less than 1

(3) More than 1

(4) Infinite

45) In case of positive deviation from Raoult's law, interaction of -

(1) A-B > A-A or B-B

(2) A-B < A-A or B-B

(3) A-B = A-A or B-B

(4) None of these

@EXAMLIX

ANSWER KEYS

## PHYSICS

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 2 | 1 | 3 | 3 | 4 | 1 | 4 | 3 | 3 | 4 | 1 | 3 | 1 | 2 | 2 | 3 | 3 | 1 | 1 | 4 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 2 | 2 | 3 | 3 | 2 | 4 | 3 | 2 | 4 | 3 | 3 | 4 | 1 | 1 | 1 | 3 | 1 | 4 | 2 | 2 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 3 | 3 | 3 | 4 | 3 | | | | | | | | | | | | | | | |

## CHEMISTRY

| Q. | 46 | 47 | 48 | 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 | 65 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 3 | 1 | 3 | 2 | 3 | 1 | 1 | 2 | 3 | 2 | 3 | 1 | 4 | 2 | 2 | 3 | 2 | 3 | 4 | 4 |
| Q. | 66 | 67 | 68 | 69 | 70 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 | 81 | 82 | 83 | 84 | 85 |
| A. | 3 | 3 | 3 | 3 | 4 | 1 | 1 | 3 | 1 | 4 | 1 | 1 | 2 | 2 | 2 | 3 | 1 | 3 | 2 | 1 |
| Q. | 86 | 87 | 88 | 89 | 90 | | | | | | | | | | | | | | | |
| A. | 4 | 4 | 2 | 2 | 2 | | | | | | | | | | | | | | | |

@EXAMLIX

SOLUTIONS

PHYSICS

1) NCERT XI, Part-I, Pg.# 60-61

$F_{net} = ma$

$ma = \mu mg$

$a = \mu g$

In the frame of belt

$u_{rel} = 3$ m/s

$v_{rel} = 0$

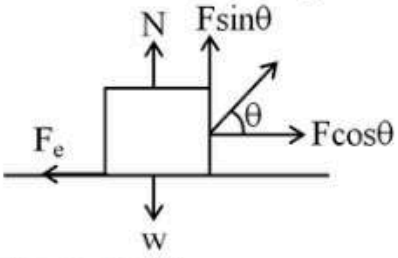
$a_{rel} = -4$ m/s$^2$

$v^2 = u^2 + 2as$

$0^2 = 3^2 + 2(-4)s$

$s = 9/8 = 1.125$ m

2) NCERT XI, Part-I, Pg.# 60-61




$N = w - F\sin\theta$

To just start motion,

$F\cos\theta = \mu N$

$F\cos\theta = \mu(w - F\sin\theta)$

$F(\cos\theta + \mu\sin\theta) = \mu w$

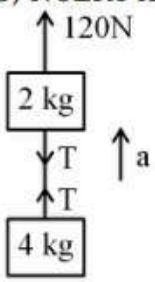
$$F = \frac{\mu W}{\cos\theta + \mu \sin\theta}$$

For $F_{min}$, $\cos\theta + \mu\sin\theta \rightarrow$ Maximum

$$F_{min} = \frac{\mu W}{\sqrt{1+\mu^2}}$$

$$(-\sqrt{1+\mu^2} \le (1\cos\theta + \mu\sin\theta) \le \sqrt{1+\mu^2})$$

3) NCERT XI, Part-I, Pg.# 53




$F_{net} = ma$

$120 - 6g = 6a$

$a = 10$ m/s$^2$

@EXAMLIX

$T - 4g = 4a$
$T = 4g + 4a$
$T = 40 + 40$
$T = 80\ N$

4) NCERT XI, Part-I, Pg.# 62
For the same weight, rolling friction is much smaller (even by 2 or 3 order of magnitude) than static or sliding friction.

5) NCERT XI, Part-I, Pg.# 58
Velocity = constant, because magnitude and direction of velocity is constant.
$\square F_{net} = ma$
$\square F_{net} = 0$

6) NCERT XI, Part-I, Pg.# 55
$x(t) = 3t + 4t^2 + 5t^3$
$v = \frac{dx}{dt} = 3 + 8t + 15t^2$
$a = \frac{dv}{dt} = 8 + 30t$
at $t = 2s$, $a = 8 + 30(2) = 68\ m/s^2$
$\square F = ma$
$= 2 \times 68 = 136\ N$

@EXAMLIX

7) NCERT XI, Part-I, Pg.# 56
For action-reaction pair it is not necessary that bodies are in physically contact with each other, for example electrostatic force between 2 charges gravitational force etc.
According to Newton's first law no force is required to maintain a uniform motion of a body or particle.

8)

NCERT XI, Part-I, Pg.# 56
According to third law of motion it is impossible to have a single force out of mutual interaction between two bodies, whether they are moving or at rest. While Newton's third law is applicable for all types of forces.

9) NCERT XI, Part-I, Pg.# 69
$F = \frac{\Delta p}{\Delta t}$ and slope of x-t graph = velocity
(A) At $t < 0 \rightarrow \Delta p = 0 \rightarrow F = 0$
(B) At $t > 4s \rightarrow \Delta p = 0 \rightarrow F = 0$
(C) At $0 < t < 4s \rightarrow \Delta p = 0 \rightarrow F = 0$
(D) At $1s \le t \le 5s \rightarrow F = \frac{m(v_2 - v_1)}{t}$

$$= \frac{4\left(0-\frac{3}{4}\right)}{5-1} \Rightarrow -\frac{3}{4} = -0.75 \text{ N}$$

10) NCERT XI, Part-I, Pg.# 61

In case-1

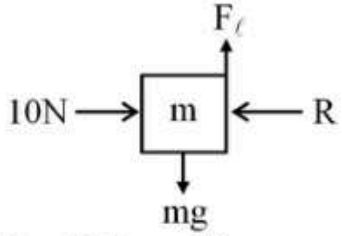
$$a = \frac{mg}{m+m} = \frac{g}{2}$$

In case-2

$a' = 2a = 2(g/2) = g$

This is a condition of free fall, which is possible if either mass placed on the table is zero or additional mass required to hang is very large as compare to mass placed on the table.

11) NCERT XI, Part-I, Pg.# 61




R = 10 N ... (i)

$F_{\square} = mg$

$\mu_s R = mg$ ... (ii)

From (i) and (ii)

$0.2 \times 10 = mg$

□ Weight = 2N

$= \frac{2}{g}$kg – wt

$= = \frac{2}{9.8} = 0.204$ kg-wt

@EXAMLIX

12) NCERT XI, Part-I, Pg.# 66

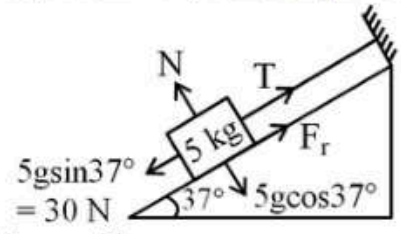
According to Newton's second law

F = ma

If F = 0, a = 0

According to Newton's first law, if external force on a body is zero then it remains at rest if initially at rest and if the body was already moving with some constant velocity then it continues moving with that same constant velocity.

13) NCERT XI, Part-I, Pg.# 61




$f_\ell = \mu_s N$

$= 0.8 \times 5g\cos 37°$

$= 0.8 \times 50 \times \frac{4}{5} = 32$ N

$f_\ell > 5g\sin 37^\circ$

 Body remains at rest.

Here friction is strong enough to keep the block from sliding on its own, the string does not need to pull it. Therefore, the tension (T) is 0 N.

14) NCERT XI, Part-I, Pg.# 55

Impulse = Change in momentum

$I = F\Delta t = \Delta p = m\Delta v$

Area under force-time graph = $m\Delta v$

For time t = 0 to t = 4s :

$\frac{1}{2}(3+4)(8) = 2(\Delta v)$

$\Delta v = 14$ m/s

15) NCERT XI, Part-I, Pg.# 60

$\mu_s = \tan\theta_{max}$

$\mu_s = \tan 15^\circ$

$= 0.27$

16)

NCERT Pg. # 57

$F = \frac{dp}{dt} = m\frac{dv}{dt}$

F = m × (Slope of v-t graph)

@EXAMLIX

17) NCERT XI, Part-I, Pg.# 53-54

$\vec{F}_{Net} = \vec{0}$

$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = \vec{0}$

$\vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = -\vec{F}_1$

acceleration $\Rightarrow \frac{-\vec{F}_1}{m}$

18) Initially spring is unstretched so tension = 0 so acceleration of both blocks = g downwards.

19)

NCERT Pg. # 55

$\Delta P$ = Area of graph F-t w.r.t. t-axis

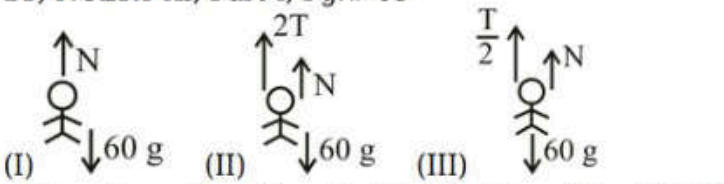
$= \frac{1}{2}[3+4] \times 20 + \left[-\frac{1}{2} \times 1 \times 20\right]$

= 60 kg m/s

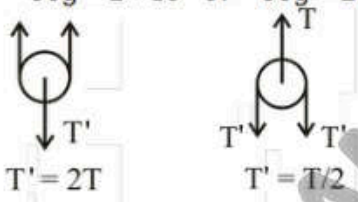
20)

NCERT XI, Part-I, Pg.# 53-55
Statement (A) : ( Correct ) Both force (F) and acceleration (a) are vector quantities, so the equation F=ma is a vector relationship.
Statement (B) : ( Correct ) The acceleration at any instant depends only on the net force acting at that instant ($a = F_{net}/m$), not on the past position or velocity. It is "local" in time.
Statement (C) : ( Incorrect ) It is applicable to rigid bodies and systems of particles. For an extended body, $F_{net}=Ma_{cm}$, where $a_{cm}$ is the acceleration of the center of mass.
Statement (D) : ( Incorrect ) The $F_{net}$ in the second law includes all external forces (both conservative like gravity/spring force and non-conservative like friction/viscous drag).
The incorrect statements are (C) and (D).

21) NCERT XI, Part-I, Pg.# 65




(I) $N = 60g$ (II) $N = 60g - 2\times40$ (III) $N = 60g - 20 = 580$ N




$T' = 2T$ $T' = T/2$

(1) $T = 40$ (II) $T' = 80$N (III) $T' = 20$N

©EXAMLIX

22) NCERT XI, Part-I, Pg.# 60-61
Max. friction depends on mass of the body too.

23) NCERT XI, Part-I, Pg.# 57-58, 66

$k = \frac{1}{2}mv^2 \Rightarrow \frac{dk}{dx} = F = \frac{dp}{dt}$

$\frac{dk}{dx} = \frac{m}{2} = 2v\frac{dv}{dx}$ (At C, Magnitude of slope is maximum)

$\frac{dk}{dx} = ma = F$

24) NCERT XI (Part-1), Pg. # 28
Collinear vectors must lie in the same plane so they must be coplanar vectors also.

25) NCERT, Pg. # 34

$|\vec{a}| = 26$

$\sqrt{10^2 + m^2} = (26)^2$

$100 + m^2 = 676$

$m^2 = 576$

$m = \pm 24$, positive value of $m = 24$

26) NCERT XI (Part-1), Pg. # 71-72

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|\,|\vec{b}|}$$

$$\cos\pi/3 = \frac{(\hat{i}+\hat{k})\cdot(\hat{i}-\hat{j}+\lambda\hat{k})}{\sqrt{1^2+1^2}\sqrt{1^2+1^2+\lambda^2}}$$

$$\frac{1}{2} = \frac{1+\lambda}{\sqrt{2}\sqrt{2+\lambda^2}}$$

By solving

$2\lambda^2 + 8\lambda = 0$

$2\lambda(\lambda + 4) = 0$

$\lambda = 0, \lambda = -4$

27) NCERT XI (Part-1), Pg. # 71-72, 102-103

If $\vec{A}\perp\vec{B} \Rightarrow \vec{A}\cdot\vec{B} = AB\cos 90° = 0$

If $\vec{A}\|\vec{B} \Rightarrow \vec{A}\times\vec{B} = AB\sin 0°\hat{n} = 0$

28) NCERT XI (Part-1), Pg. # 33

$$\frac{\cos^2\alpha + \cos^2\beta + \cos^2\gamma}{\sin^2\alpha + \sin^2\beta + \sin^2\gamma}$$

$$= \frac{\cos^2\alpha + \cos^2\beta + \cos^2\gamma}{(1-\cos^2\alpha)+(1-\cos^2\beta)+(1-\cos^2\gamma)}$$

$$\Rightarrow \frac{\cos^2\alpha + \cos^2\beta + \cos^2\gamma}{3-(\cos^2\alpha+\cos^2\beta+\cos^2\gamma)} = \frac{1}{3-1} = \frac{1}{2}$$

@EXAMLIX

29) NCERT XI (Part-1), Pg. # 71-72

$\vec{A}\perp\vec{B}$, $\vec{A}\cdot\vec{B} = 0$

$(2\hat{i} + 4\hat{j})\cdot(p\hat{i} + q\hat{j}) = 0$

$2p + 4q = 0$

$\frac{p}{q} = -2$

30) NCERT XI (Part-1), Pg. # 34

Let $\vec{R} = \vec{P} + \vec{Q}$

Then magnitude of $\vec{R}$ : $\sqrt{P^2 + Q^2 + 2PQ\cos\theta}$

$R_{min} = |\vec{P}| - |\vec{Q}|$ (at $\theta = 180°$)

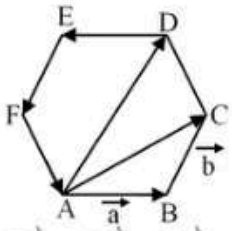
$R_{max} = |\vec{P}| + |\vec{Q}|$ (at $\theta = 0°$)

$\therefore$ Range : $|\vec{P}| - |\vec{Q}| \leqslant |\vec{R}| \leqslant |\vec{P}| + |\vec{Q}|$

31) NCERT XI (Part-1), Pg. # 28-29

$\overrightarrow{AD} = 2\overrightarrow{BC} = 2\vec{b}$

In $\Delta ACD$




$\vec{AD} = \vec{AC} + \vec{CD}$

$\therefore \vec{CD} = \vec{AD} - \vec{AC}$

$= 2\vec{b} - (\vec{a} + \vec{b})$

$= \vec{b} - \vec{a}$

32) NCERT XI (Part-1), Pg. # 71-72, 102-103

(1) $\vec{A} \cdot \vec{B} = (\hat{i} - \hat{k}) \cdot (\hat{i} + \hat{j}) = \hat{i} \cdot \hat{i} + \hat{i} \cdot \hat{j} - \hat{k} \cdot \hat{i} - \hat{k} \cdot \hat{j}$

$= 1 + 0 - 0 - 0 = 1$

(2) $\vec{B} - \vec{A} = (\hat{i} + \hat{j}) - (\hat{i} - \hat{k}) = \hat{j} + \hat{k}$

(3) $\cos\theta = \dfrac{\vec{A} \cdot \vec{B}}{AB} = \dfrac{(\hat{i} - \hat{k}) \cdot (\hat{i} + \hat{j})}{\sqrt{2}\sqrt{2}} \Rightarrow \dfrac{1}{2} = 0.5$

(4) $|\vec{A} + \vec{B}| = |2\hat{i} + \hat{j} - \hat{k}| = \sqrt{2^2 + 1^2 + 1^2} \Rightarrow \sqrt{6}$

33) NCERT XI (Part-1), Pg. # 46

Magnitude of a vector is always a scalar.

For eg. $\vec{F} = -2N\hat{i}$

$|\vec{F}| = \dfrac{\vec{F}}{\hat{F}} \Rightarrow \dfrac{-2N\hat{i}}{-\hat{i}} = 2N$

Each component of a vector is always a vector.

@EXAMLIX

34) NCERT XI (Part-1), Pg. # 29

Both (A) and (R) are true but (R) is not correct explanation of (A).

Range of resultant of two vectors

$A - B \leqslant |\vec{A} + \vec{B}| \leqslant A + B$

35)

NCERT XI (Part-1), Pg. # 29

Both Statement I and Statement II are wrong.

(1) Direction of $\lambda\vec{A}$ is opposite of $\vec{A}$ if $\lambda < 0$ and same as $\vec{A}$ if $\lambda > 0$.

(2) If a vector is multiplied by zero, then resultant is a is a null vector.

36) NCERT, Pg. # 29

Direction of $2\vec{a}$ : Same as $\vec{a}$

Direction of $-3\vec{b}$ : Opposite to $\vec{b}$

$\therefore$ Angle between $2\vec{a}$ and $-3\vec{b}$ is = $\pi - \theta$

$= \pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3}$

37) NCERT XI (Part-1), Pg. # 32

Projection of $\vec{b}$ along $\vec{a}$ = b cosθ (θ is angle between $\vec{a}$ & $\vec{b}$ Projection of $\vec{a}$ along $\vec{b}$ = a cos θ (θ is angle between $\vec{a}$&$\vec{b}$)

Projection of $\vec{b}$ along x-axis = b $\cos\theta_x$ ($\theta_x$ is angle between $\vec{b}$ & x-axis)

Projection of $\vec{a}$ along y-axis = acos $\theta_y$ = acos$\theta_y$ ($\theta_y$ is angle between $\vec{a}$ & y-axis)

38) NCERT XI (Part-1), Pg. # 71-72, 102-103

$|\vec{b} \times \vec{c}| = \sqrt{15}$

$bc \sin\theta = \sqrt{15} \Rightarrow (4)(1)\sin\theta = \sqrt{15}$

$\sin\theta = \frac{\sqrt{15}}{4}$

$\cos\theta = \sqrt{1-\sin^2\theta}$

$= \sqrt{1-\frac{15}{16}} = \pm\frac{1}{4}$

Now $\vec{b} - 2\vec{c} = \lambda\vec{\alpha}$

By self dot product

$(\vec{b}-2\vec{c})\cdot(\vec{b}-2\vec{c}) = (\lambda\vec{a})\cdot(\lambda\vec{a})$

$b^2 - 4bc\cos\theta + 4c^2 = \lambda^2 a^2$

Case (1): If $\cos\theta = +\frac{1}{4}$

$(4)^2 - 4(4)(1)\times\frac{1}{4} + 4(1)^2 = \lambda^2(1)^2$

$\lambda^2 = 16$

$\boxed{\lambda = \pm 4}$

Case (2) : If $\cos\theta = -\frac{1}{4}$

$(4)^2 - 4(4)(1)\times\left(-\frac{1}{4}\right) + 4(1)^2 = \lambda^2(1)^2$

$16 + 4 + 4 = \lambda^2$

$\boxed{\lambda = \pm\sqrt{24}}$

From the given options, option (4) is correct.

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39)

Ref. - Pg. # 46

$\vec{V} = |\vec{V}|\hat{V} = 10\frac{(-4\hat{i}+3\hat{j})}{5}$

$= -8\hat{i} + 6\hat{j}$ m/s

40)

NCERT-XI, Part-I, Pg. # 34

$\left|\hat{a}-\hat{b}\right| = \sqrt{(1)^2+(1)^2-2\times1\times1\cos60^\circ}$

$\left|\hat{a}-\hat{b}\right| = \sqrt{(1)^2+(1)^2-2\times1\times1\times\frac{1}{2}} = 1$

41)

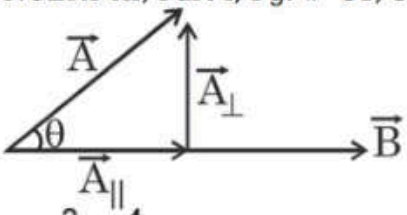
NCERT-XI, Part-I, Pg. # 34
Change in velocity = $2v\sin(\theta/2) = 2v\sin 20^\circ$

42)

NCERT-XI, Part-I, Pg. # 28, 29
If two vector are parallel
Then $= \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$
so here $\frac{1}{2} = \frac{1}{2} = \frac{-1}{\lambda}$
$\lambda = -2$

43)

NCERT-XI, Part-I, Pg. # 31, 32




$\hat{B} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$

$\vec{A}_{||} = (A\cos\theta)\hat{B} = (\vec{A}\cdot\hat{B})\hat{B} = \frac{21}{25}\hat{i} + \frac{28}{25}\hat{j}$

$\vec{A}_{\perp} = \vec{A} - \vec{A}_{||} = (\hat{i} + \hat{j} + 5\hat{k}) - \left(\frac{21}{25}\hat{i} + \frac{28}{25}\hat{j}\right)\frac{4}{25}\hat{i} - \frac{3}{25}\hat{j} + 5\hat{k}$

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44) NCERT-XI, Part-I, Pg. # 102, 103
$\vec{A}\times\vec{B}$ is a vector perpendicular to $\vec{A}, \vec{B}\ \vec{A}+\vec{B},\ \vec{A}-\vec{B}$ But $\vec{A}\cdot\vec{B}$ is a scalar quantity

45)

NCERT-XI, Part-I, Pg. # 34
As, $\vec{P} = \vec{F}_1 + \vec{F}_2$

- $P^2 = F_1^2 + F_2^2 + 2F_1F_2\cos\theta$
- $\vec{Q} = \vec{F}_1 - \vec{F}_2$
- $Q^2 = F_1^2 + F_2^2 - 2F_1F_2\cos\theta$
- $P^2 + Q^2 = 2(F_1^2 + F_2^2)$

$= 2(18 + 32) = 100$

$\sqrt{P^2 + Q^2} = \sqrt{100} = 10$ N

CHEMISTRY

46)

**Explain Question:** Identify type of solution for Cu in Au

**Concept:** Types of solutions

**Solution:**
Copper and gold both are **solids**
When one solid dissolves in another → **solid in solid solution (alloy)**

**Final Answer:** Option (3)

47) $\frac{P_1^o - P_S}{P_1^o} = \frac{n_B}{n_A + n_B}$
For very dilute solution;
$\frac{P_1^o - P_S}{P_1^o} = \frac{n_B}{n_A}$
$\frac{P_1^o - P_S}{P_1^o} = \frac{w_2 \times M_1}{M_2 \times w_1}$

48) NCERT-XII, Pg. No. # 49

49) NCERT-XII, Pg. No. # 52

50) NCERT-XII, Pg. No. # 51

51) NCERT-XII, Pg. No. # 53

52) NCERT-XII, Pg. No. # 53

53) $\Delta T_f = iK_f \times m$

54) $\frac{P^o - P_S}{P^o} = x_B$

$= \frac{1.013 - 1.004}{1.013} = \frac{n_B}{n_A + n_B}$

$= 0.0088 = \frac{n_B}{n_A}$

$= 0.0088 = \frac{\frac{2}{x}}{\frac{98}{18}}$

$= 0.0088 = \frac{36}{98x} \Rightarrow x = 41.37$

55) Theory based

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56) **Explain the Question :** To find out mole fraction.
**Concept :** This question is based on mole fraction.
**Solution :**

Benzene ⇒ w = 39g  moles = $\frac{39}{78} = 0.5$

ethyl alcohol ⇒ w = 46g  moles = $\frac{46}{46} = 1$

$X_B = \frac{n_B}{n_T} = \frac{0.5}{1.5} = \frac{1}{3}$

$= 0.33$

**Final answer :** Option (3)

57) $m = \frac{2.5 \times 1000}{60 \times 75}$

58) Solubility of solids in liquids are not affected by change in pressure.

59) Solubility of gas in liquid is exothermic.

60) $K_H \propto \frac{1}{x_{gas}}$

61) Fact Based.

62)

**Question Explanation :** We need to find the boiling point of a glucose solution. First calculate $\Delta T_b$ using molality, then add to water's boiling point (373 K).

**Concept :** $\Delta T_b = K_b \times m$
Molality m = moles of solute / kg solvent

**Solution :**

Moles of glucose = $\frac{18\ g}{180\ g}\ mol^{-1} = 0.1\ mol$
Solvent = 1 kg water → molality m = 0.1 mol $kg^{-1}$

$\Delta T_b = K_b \times m = 0.52 \times 0.1 =$ **0.052 K**

Boiling point = 373 + 0.052 = 373.052 K ≈ 373.05 K

**Final Answer :** Option (2)

63)

**Explain Question:** Identify process used for desalination of sea water

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**Concept:** Reverse osmosis

**Solution:**
In desalination, **pressure is applied on solution side** to force solvent to move opposite to osmosis
This process is called **reverse osmosis**, used to remove salts from seawater

**Final Answer:** Option (3)

64) B.P $\propto$ i

65) $H_2O$ & $HNO_3$ is azeotropic mixture

66) $M_1V_1 = M_2V_2$

67) $i_{urea} = 1$
$i_{CaBr_2} = 3$

68) i = 1 + (4 – 1) . 5 = 2.5

69) **Concept :** At high pressure, more nitrogen dissolves in the blood. During rapid ascent, this excess nitrogen forms bubbles in the blood and tissues.
**Solution :** Bends, also known as decompression sickness, occur due to the toxic effect of the high concentration of **Nitrogen in blood**.
**OR**
NCERT, Part-1, Chapter-Solution, Topic-Application of Henry's Law

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70)

**Question:** Select the true statement regarding the properties of cellulose acetate.

**Concept:** Reverse Osmosis

**Solution:** Cellulose acetate is a semi-permeable material that allows water molecules to pass through while blocking salts and impurities. Because of these specific characteristics, it is the standard membrane used in desalination.

**Final Answer:** Option (4)

71) For ideal solution
$\Delta H_{Mix} = 0; \Delta V_{Mix} = 0$

72) $\Delta T_f = K_f m \times i$
$2k_f = k_f \times m \times 2$ $\boxed{m = 1}$

$$\frac{p^0 - p_s}{p_s} = \frac{im \times M_{solvent}}{1000} = \frac{2 \times 18}{1000} = .036$$

73)

**Explain Question :** We need the freezing point of a 0.5 m aqueous NaBr solution that is 80% dissociated.
**Concept :** This question is based on
Freezing point depression: $\Delta T_f = i.K_f.m$
Van't Hoff factor: $i = 1 + \alpha(n - 1)$ where n = number of ions produced.
For $NaBr \rightarrow Na + Br^-$, n = 2
Degree of dissociation $\alpha = 0.8$
**Solution :**
$i = 1 + 0.8(2 - 1) = 1.8$
$\Delta T_f = i.K_f.m = 1.8 \times 1.86 \times 0.5$
$\Delta T_f = 1.674K$
Normal freezing point of water = 273K
New freezing point = 273 - 1.674 ≈ 271.32K .
**Final answer :** 3

74) $$\Delta T_f = \frac{K_f \times W_2 \times 1000}{M_2 \times W_1}$$
$$W_2 = \frac{1.5 \times 1.76 \times 75}{3.9 \times 1000}$$
≅ 5.08 gm

75) No of moles of $Na^+$ = 92/23 = 4
$$m = \frac{n_{solute}}{w_{solvent(kg)}} = \frac{4}{1} = 4$$

76) $P = K_H \times x$
$$x = \frac{P}{K_H} = \frac{760}{4.27 \times 10^5} \cong 178 \times 10^{-5}$$

77)

$$KCl \rightarrow \xrightarrow[\text{conc.}\downarrow]{0.3\ M} \xrightarrow[=\alpha\uparrow]{0.2\ M} \xrightarrow[=i\uparrow]{0.2\ M}$$

78)

**Explain Question :** A 1 molal solution of a carboxylic acid in benzene shows $\Delta T_b = 1.518$ K. Using $K_b = 2.53$ K kg $mol^{-1}$, find the degree of association ($\beta$) for dimerization of the acid in benzene.
**Concept :** This question is based on

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**For association:** $\beta = \frac{1-i}{1-1/n}$

$i = \frac{\Delta T_b}{K_b.m}$

**Solution :**

**Given :** $m = 1$, $\Delta T_b = 1.518$, $K_b = 2.53$

$$i = \frac{(\Delta T_b)_{obs}}{(\Delta T_b)_{normal}} = \frac{1.518}{2.53 \times 1} = .6$$

$$\beta = \frac{1-i}{1-1/n}$$

$$= \frac{1-.6}{1-1/2} = \frac{.4}{.5} = .8 \text{ or } 80\%$$

**Final answer :** 2

79)

**Explain Question :** Van't Hoff factor depends on dissociation/association; here $Hg_2Cl_2$ dissociates into 3 ions ($Hg_2^{2+}$ + $2Cl^-$).

**Concept :** This question is based on Van't Hoff factor for dissociation → $i = 1 + \alpha(n - 1)$

**Solution :**

$Hg_2Cl_2 \rightarrow Hg_2^{2+} + 2Cl^- \rightarrow n = 3$

$\alpha = 0.8$

$i = 1 + 0.8(3 - 1) = 1 + 1.6 = 2.6$

**Final answer :** 2

80) $\left(\frac{18}{180} + \frac{6}{60}\right) \times 0.0821 \times 300 = \pi \times 1$

$\boxed{\pi = 4.926 \text{ atm}}$

81) $\Delta T_f = i\ k_f m$

$7.1 \times 10^{-3} = i \times 1.86 \times .001$

$\boxed{i = 3.817}$

$$\alpha = \frac{i-1}{n-1}$$

$$i = \frac{3.87-1}{(x+1)-1}$$

$$x = 2.817 \cong 3$$

82) $P_T = P_A^o x_A + P_B^o x_B$

$760 = 520x_A + 1000(1-x_A)$

$\boxed{x_A = .5}$ mol % = 50

83)

$P_s = p^0_A X_A + p^0_B X_B$

$600 = 450(1-X_B) + 700 + X_B$

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84) $\frac{\Delta T_b = K_b m}{\Delta T_f = K_f m}$

$\frac{.17}{\Delta T_f} = \frac{.512}{1.86}$

$\Delta T_f = \frac{0.17 \times 1.86}{.512} = 0.62^{\circ}C$

F.P = 0. - $\Delta T_f$ = 0 - .62

= -.62°C

85) $CCl_4$ and $C_6H_6$ form non ideal solution (+ve) deriation

86) Mass of solution = 100 gm

Mass of NaOH = 10 gm

Mass of solvent = 90 gm

Molality = $\frac{10/40}{90/1000} = 2.778$

87) $\pi = CRT$

(i) $\pi = RT$

(ii) $\pi = RT$

(iii) $\pi = RT$ (mixture has conc = 1 M)

88) $M = \frac{0.5 \times 1000}{32 \times 1}$

$= \frac{50}{32}$

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89) Fact Based

90) NCERT, Part-1, Chapter-Solution, Topic-Non-ideal solution